FLAT BASE CHANGE FOR ADEQUATE FUNCTORS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file adequate.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans are self-contained: no reference leaves them and no bibliography citation occurs in them.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The presentation of an adequate functor as a kernel of a map of module-valued functors. The definition of adequate functors, invoked at proof line 450. Supplies the exact sequence used in the diagram.
- Flatness of B -> B' makes the tensor product functor exact. Standard property of flat ring maps, used at proof line 462. Makes the top row of the diagram exact so that the five lemma applies.

Lemma 0.1. *Let A be a ring. Let F be an adequate functor on Alg_A. If $B \to B'$ is flat, then $F(B) \otimes_B B' \to F(B')$ is an isomorphism.*

Proof. Choose an exact sequence $0 \to F \to \underline{M} \to \underline{N}$. This gives the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & F(B) \otimes_B B' & \longrightarrow & (M \otimes_A B) \otimes_B B' & \longrightarrow & (N \otimes_A B) \otimes_B B' \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & F(B') & \longrightarrow & M \otimes_A B' & \longrightarrow & N \otimes_A B' \end{array}$$

where the rows are exact (the top one because $B \to B'$ is flat). Since the right two vertical arrows are isomorphisms, so is the left one. □

Date: stacks-project:adequate.tex:local-snapshot-20261012; GNU FDL 1.2 or later.